

Figures of Section 5 (theory) — source collection

Each figure below is the same TikZ/pgfplots code as in `theory_section.tex`; copy the corresponding `\begin{figure}...``\end{figure}` block (and the colour/magnet definitions from the preamble of this file). Figure 13 is the external file `fig_release.pdf`, produced by `make_fig_release.py`. House style: springs copper (= `orange!80!black`), clamps `gray!60`, base `gray!30`, magnets S = `red!70`, N = `blue!70`, exactly as in the setup figure of Section 1.

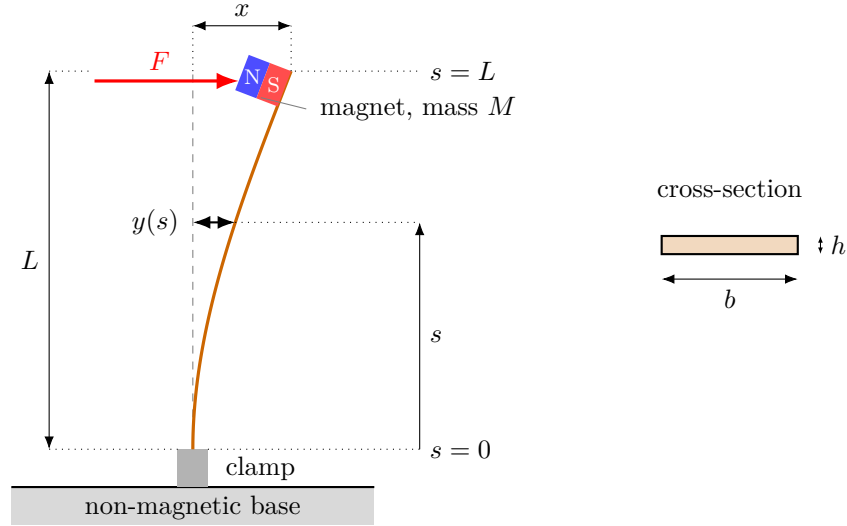


Figure 1: Geometry of the leaf spring.

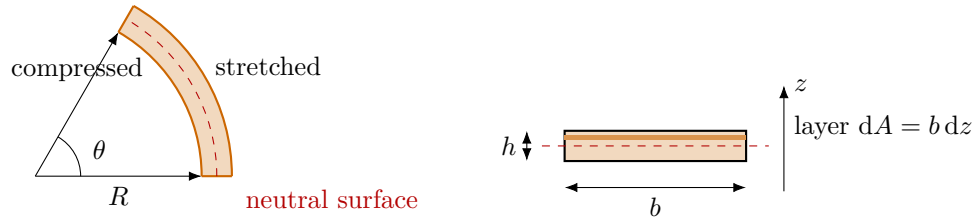


Figure 2: Left: a short element of the strip bent to radius of curvature R through angle θ . Layers outside the neutral surface (dashed) are stretched, layers inside are compressed, by a strain z/R . Right: the cross-section, with z measured from the neutral surface; a layer of area $dA = b dz$ at height z carries stress Ez/R , and the sum of their moments is EI/R .

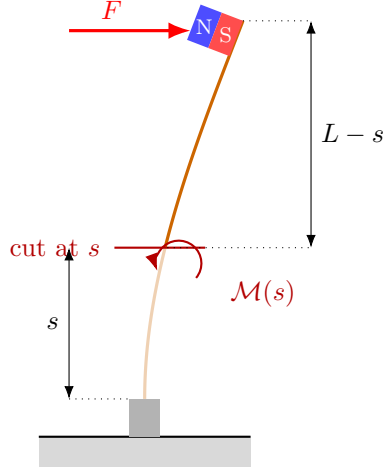


Figure 3: Finding the bending moment. The part of the strip above an imaginary cut at height s (full colour, together with the magnet) is acted on by the tip force F , with lever arm $L - s$, and by the internal moment $\mathcal{M}(s)$ exerted on it by the part below the cut (faint). For equilibrium of the upper part, $\mathcal{M}(s) = F(L - s)$: largest at the clamp, zero at the tip.

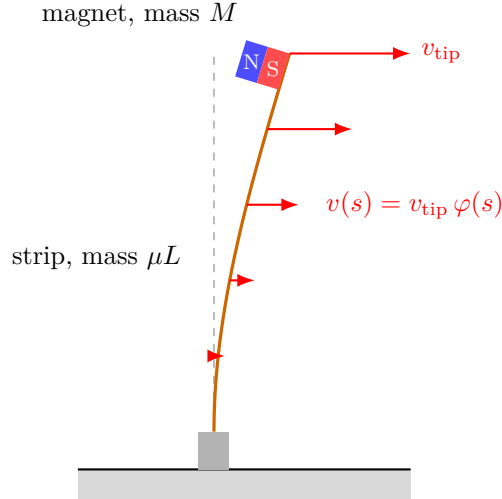


Figure 4: Rayleigh's method. During the oscillation every point of the strip is assumed to move in proportion to the static deflection curve $\varphi(s)$ of equation (9), so its velocity is $v_{\text{tip}}\varphi(s)$: nearly zero near the clamp, equal to the magnet's velocity at the tip (arrows). The strip's kinetic energy is the same as that of a point mass $\frac{33}{140}\mu L$ moving with the tip.

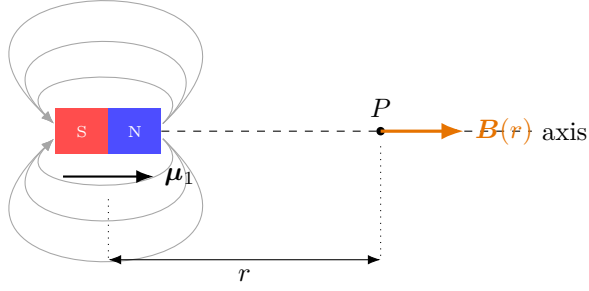


Figure 5: The field of a small magnet (a magnetic dipole of moment μ_1 , magnitude μ_m , pointing from S to N). Grey: field lines. At a point P on the axis, at distance r from the *centre* of the magnet, the field points along the axis in the direction of the moment, with magnitude $\mu_0 2\mu_m / 4\pi r^3$.

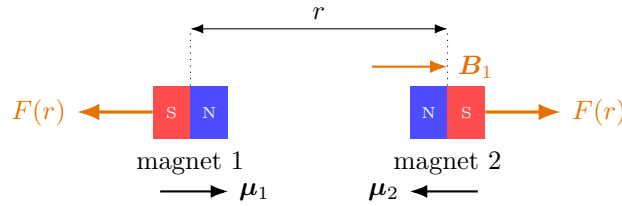


Figure 6: Two identical magnets on a common axis with like poles (N) facing. The field B_1 of magnet 1 at magnet 2 points to the right; the moment μ_2 of magnet 2 points to the left, so the energy $-\mu_2 \cdot B_1$ is positive and decreases with r : the magnets repel with $F(r) = 3\mu_0\mu_m^2/2\pi r^4$, equal and opposite on the two magnets.

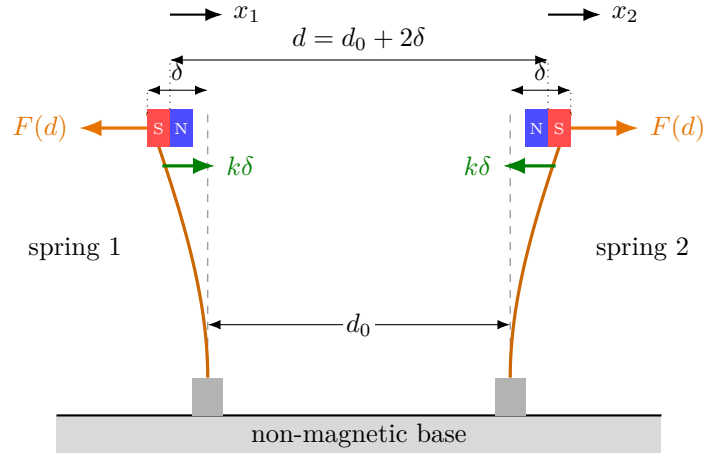


Figure 7: Equilibrium with both magnets mounted. The repulsion $F(d)$ bends each spring outwards by δ from its straight (dashed) position until the elastic restoring force $k\delta$ balances it. The centre-to-centre separation d is defined and measured in this state (d_0 is the spacing of the clamps). The displacements x_1 , x_2 used in the equations of motion are measured from these equilibrium positions, both positive to the right.

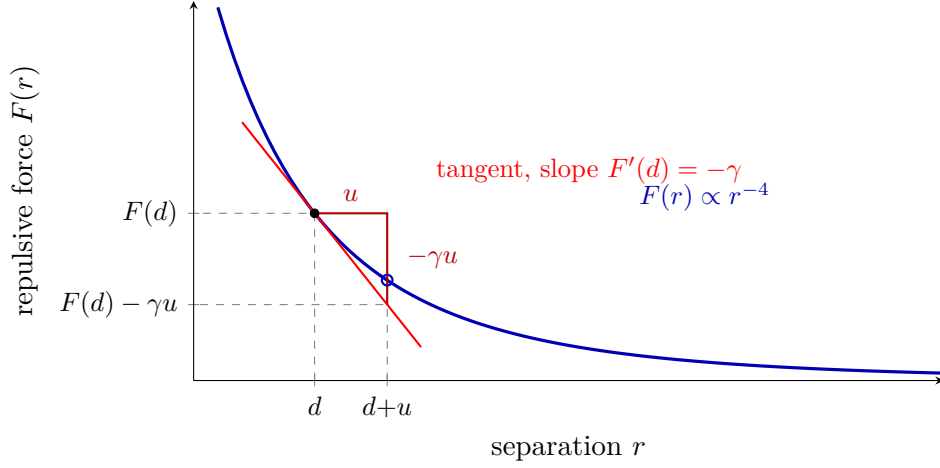


Figure 8: Linearisation of the force law. Near the equilibrium separation d the curve $F(r)$ is replaced by its tangent, whose slope is $-\gamma$. Increasing the separation by u reduces the repulsion by γu ; decreasing it increases the repulsion by the same amount—the behaviour of a compressed spring of stiffness γ . The gap between the tangent and the true curve (open circle) is the error of the approximation; it grows with u/d and is discussed in the evaluation.

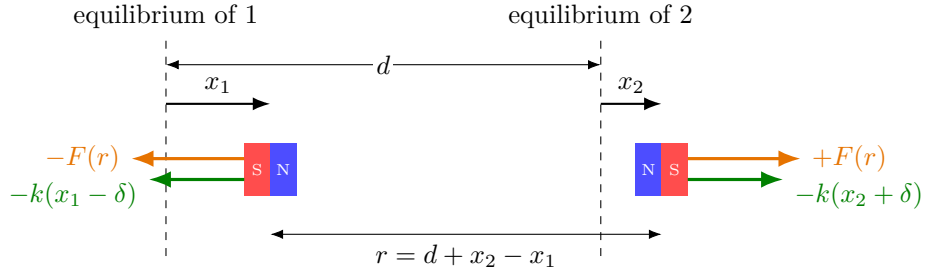


Figure 9: Free-body diagram of the two magnets (horizontal forces only, seen from above; orange: magnetic, green: elastic), both shown displaced to the right of their equilibrium positions. Each spring force is proportional to the displacement from that spring’s *unstressed* position ($+\delta$ for spring 1, $-\delta$ for spring 2, Figure 7). The magnetic force has magnitude $F(r)$ at the *current* separation and pushes the magnets apart.

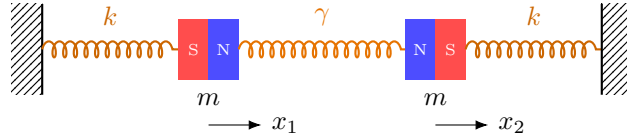


Figure 10: Mechanical equivalent of equations (23): two equal masses m on equal springs k (copper: the leaf springs), coupled by a spring of stiffness γ (orange: the magnetic repulsion). The picture is schematic; in the apparatus the coupling “spring” is the field between the magnets.

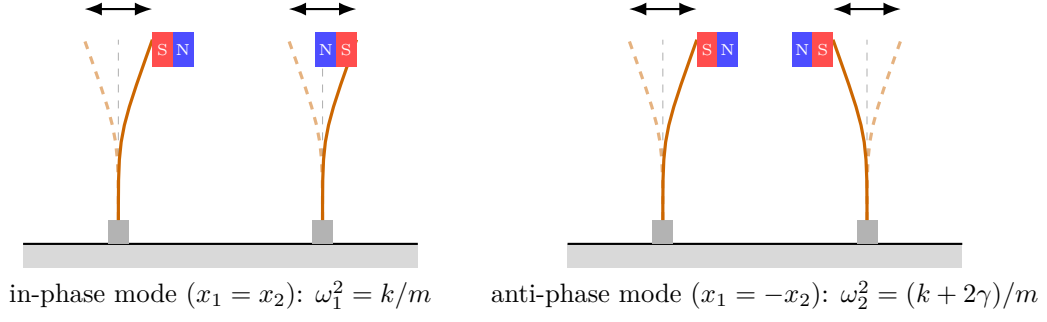


Figure 11: The two normal modes (solid: one extreme of the swing; dashed: the other; magnets drawn at the solid extreme). Left, in-phase: the magnets keep their separation, the magnetic force never changes from its equilibrium value, and each spring oscillates as if alone, at $\omega_1 = \sqrt{k/m}$. Right, anti-phase: when each magnet moves a towards the other the separation shrinks by $2a$, so each magnet feels an extra restoring force $\gamma \cdot 2a$ on top of ka ; the effective stiffness is $k + 2\gamma$.

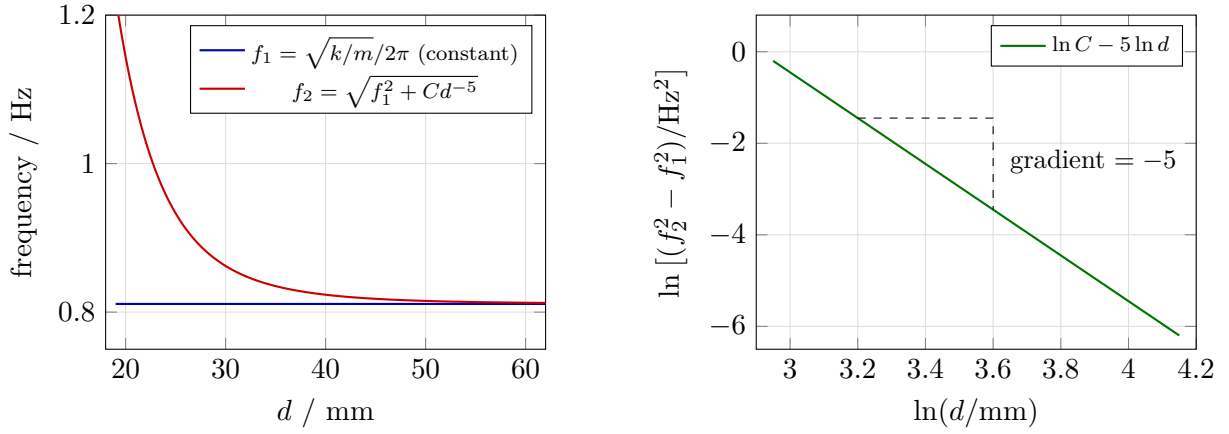


Figure 12: The prediction of the model, drawn with $f_1 = 0.811$ Hz and the value of C obtained from the data. Left: f_1 is independent of d while f_2 falls towards it. Right: the coupling term $f_2^2 - f_1^2$ on logarithmic axes is a straight line of gradient exactly -5 .

Same $f_1 = 0.802$ Hz, $f_2 = 0.996$ Hz in both rows; only the starting condition differs (linear model, $A = 8$ mm)

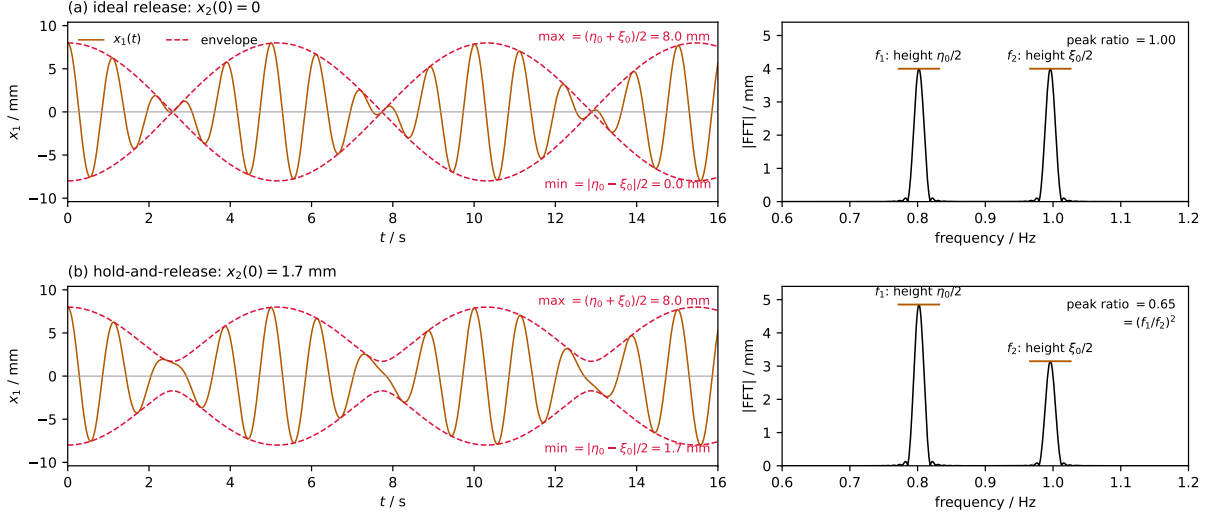


Figure 13: The same system ($f_1 = 0.802$, $f_2 = 0.996$ Hz; linear model, $A = 8$ mm) started in two ways; left: $x_1(t)$ with its envelope, right: its amplitude spectrum, with the predicted peak heights $\eta_0/2$ and $\xi_0/2$ marked. Top: idealised instantaneous release with $x_2(0) = 0$ —equal mode amplitudes, equal FFT peaks, beat minima at zero. Bottom: hold-and-release, $x_2(0) = A\gamma/(k+\gamma) = 1.7$ mm—peak ratio $(f_1/f_2)^2 = 0.65$ and beat minima of 1.7 mm. The peak *positions* are identical in both cases. (Generated by `make_fig_release.py`.)

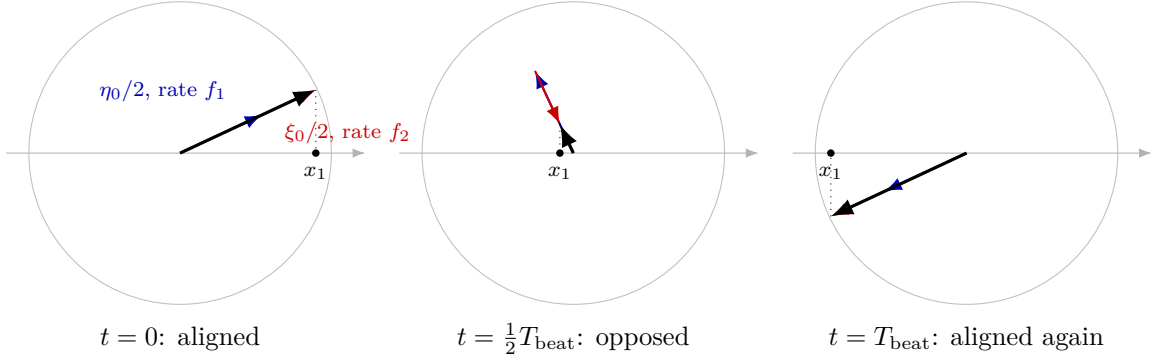


Figure 14: Beats as rotating arrows (phasors). Blue: length $\eta_0/2$, turning at f_1 revolutions per second. Red, drawn head-to-tail: length $\xi_0/2$, turning at f_2 . Black: their sum; x_1 is its horizontal projection. The relative angle grows by one full turn every $1/(f_2 - f_1)$ seconds, so the length of the black arrow—the amplitude of the fast oscillation—swings between $(\eta_0 + \xi_0)/2$ and $|\eta_0 - \xi_0|/2$ with exactly that period. Drawn for $\xi_0/\eta_0 = 0.65$, the hold-and-release case: the amplitude never falls to zero.

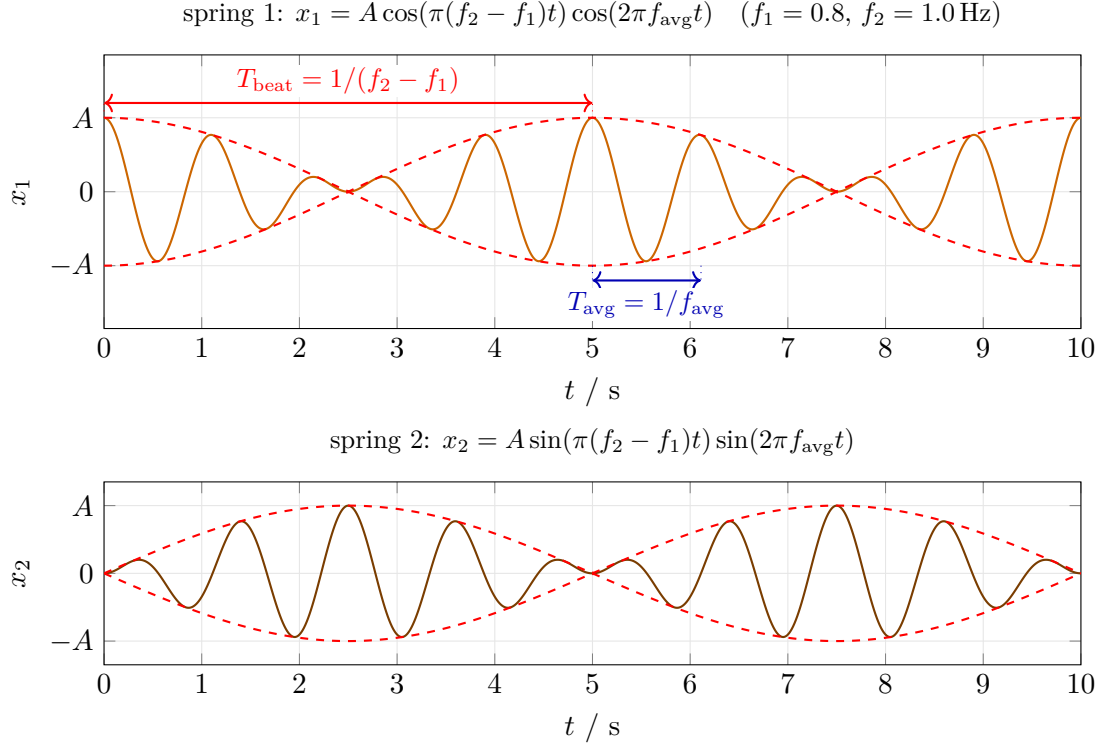


Figure 15: Beats for equal mode amplitudes. Top: the released spring and its envelope (red dashed); the swellings recur every $T_{\text{beat}} = 1/(f_2 - f_1)$, and the fast oscillation has period $1/f_{\text{avg}}$. Bottom: spring 2 carries the same two frequencies with the anti-phase term reversed, $x_2 = \frac{A}{2}(\cos \omega_1 t - \cos \omega_2 t) = A \sin(\pi(f_2 - f_1)t) \sin(2\pi f_{\text{avg}}t)$; its envelope is a quarter of a beat behind, so it swings most when spring 1 is momentarily still. This is the energy transfer between the springs.